Assignment 3: MVP Implementation & Demo

Prerequisite: Assignment 2 (Specs & MVP Definition) completed and approved.

1. Overview

In Assignment 2, you defined your "Narrow Vertical Slice"—the single most technically challenging aspect of your project. The goal of Assignment 3 is to build it.

By the deadline, you must have a functional, running prototype that demonstrates your core graphics pillars. This is not the final polished game, but it is the proof that your engineering approach works.

Crucial Rule: If your MVP definition in Assignment 2 was "A functional fluid simulation," and you submit a project with great UI but *no* fluid simulation, you will fail this assignment. Focus entirely on the core technical challenge.

2. Phase 1: Development & AI Integration

Implement the features defined in your Technical Specification.

Core Graphics Implementation

- Initialize your Game Engine/Framework.
- Implement the core algorithms (e.g., the Boids algorithm, the shader code, or the physics integration).
- Ensure the project runs at the target frame rate defined in your specs.

AI Integration Checkpoint

- **Offline AI:** If you proposed using AI for assets (textures, meshes), those assets must be present in this build.
- **Runtime AI:** If you proposed using AI for logic (LLMs, dynamic pathfinding), the API calls must be functional, even if the behavior is basic.

3. Phase 2: The Demo Video

Video Requirements (Duration: 2-3 minutes):

1. **Visual Proof:** Screen capture of the software running in real-time.
2. **Feature Walkthrough:** Voiceover or text overlay explaining exactly what is happening on

screen (e.g., *"Here you can see the shader reacting to the light source..."*).

3. **Stress Test:** Show the limits of your MVP (e.g., spawn max particles until FPS drops) to demonstrate performance characteristics.

4. Phase 3: Deliverables

Write a brief report (1-2 page PDF) analyzing your progress. It should contain the following:

Text content

- **Plan vs. Reality:**
  - What features from your Assignment 2 MVP definition made it in?
  - What features had to be cut or changed?
- **AI Utility:**
  - How useful were the AI tools you listed in your Tech Spec?
  - Did they save time, or did debugging them take longer than expected?
- **Roadmap to Final:**
  - What is left to build for the final submission?

Playable Build / Web Link:

- For Web-based: A live URL (GitHub Pages, Vercel, etc.).
- For Desktop: a link to a drive/itch.io upload. It should include a compiled executable (.exe, .app) or strict instructions on how to run.

Video Link

- YouTube or Vimeo link.

Source Code Link

- Link to a public GitHub repository. *Must include a README.md with build instructions.*

Please note that you do not need to beautify your report using LLMs. Reports should be clear, concise and comprehensively reflect your effort on the implementation. It should perform as a tool to help us understand your idea and code.